

Argument Ellipsis or More? Adjunct Recoverability in Turkish Object Gaps

Yasemin Fincan, The Ohio State University

Introduction

Ellipsis in syntax refers to the omission of material that is understood from context. Two types of ellipsis have received sustained attention in recent years: argument ellipsis (AE) and verb-stranding verb phrase ellipsis (VSVPE). In AE, an argument such as a subject or object is omitted while the rest of the clause remains intact. In VSVPE, by contrast, an entire verb phrase is elided after the verb has raised out of it, leaving the verb stranded but pronounced. On the surface, both mechanisms can yield similar strings, but their underlying structures differ in crucial ways.

To illustrate, consider Hebrew object gaps. In (1), the second clause may be analyzed as AE.

(1) a. Gil hizmin et axot-o. Yosi gam hizmin ..

Gil invite.PST ACC sister-3SG.M Yosi also invite.PST ..

‘Gil invited his sister. Yosi did too.’

b. *AE representation*

$[_{TP} Yosi_i [_{T'} gam [_{T'} hizmin_j -v -T [_{VP} t_i [_{V'} t_j [_{VP} t_j [_{DP} et\text{-}axot\text{-}o]]]]]]]$

(Adapted from Landau, 2018)

Alternatively, the same string can be analyzed as VSVPE, where the verb raises out of VP and the VP remnant is deleted. The stranded verb surfaces overtly, as in (2).

- (2) a. Gil hizmin et axot-o. Yosi gam hizmin ..
 Gil invite.PST ACC sister-3SG.M Yosi also invite.PST ..

‘Gil invited his sister. Yosi did too.’

- b. *VSVPE representation*

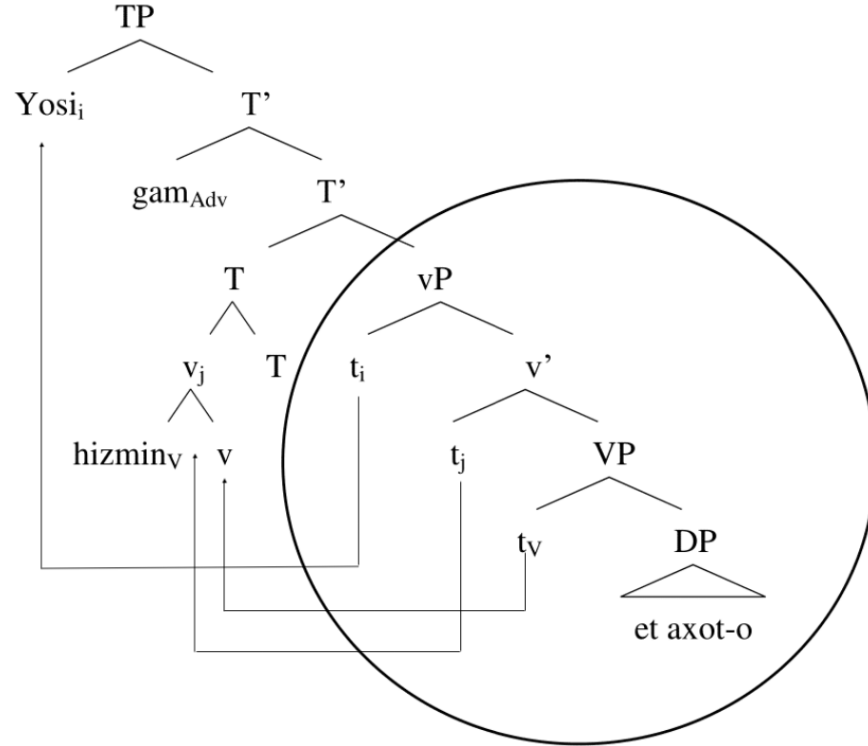


Figure 1: VSVPE representation of Hebrew: verb moves out of the ellipsis site, stranding *hizmin* ‘invite’.

(Adapted from Landau, 2018)

The crucial diagnostic for distinguishing AE from VSVPE is whether adjuncts in the antecedent clause can be interpreted inside the ellipsis site. This test, first proposed by Park (1997) and Oku (1998), has since become central in debates over ellipsis mechanisms (Funakoshi, 2016; Gribanova, 2013; Landau, 2018, 2020, 2023; Manetta, 2019; Şener & Takahashi, 2010; Simpson, 2023). If adjuncts can be recovered, this supports VSVPE; if not, AE suffices.

Turkish object gaps provide a new testing ground for this diagnostic. Consider the examples in (3).

- (3) a. Ali kek-i tarif-e göre yap-mış. Ayşe yap-ma-mış.
 Ali cake-ACC recipe-according to make-PST.EVID Ayşe make-NEG-PST.EVID
 ‘Ali made the cake according to the recipe. Ayşe didn’t (make).’
- b. Ali araba-yı hızlı yıka-mış. Ayşe yıka-ma-mış.
 Ali car-ACC fast wash-PST.EVID Ayşe wash-NEG-PST.EVID
 ‘Ali washed the car quickly. Ayşe did not (wash).’
- c. Ali makale-yi hızlı yaz-mış. Ayşe yaz-ma-mış.
 Ali article-ACC fast write-PST.EVID Ayşe write-NEG-PST.EVID
 ‘Ali wrote the article quickly. Ayşe did not (write).’

In these examples, the object in the second clause is missing, and the interpretive possibilities vary depending on the verb and the adjunct. According to the standard AE-only account for Turkish, argued by Şener and Takahashi (2010), none of these cases should allow an adjunct-inclusive reading. Yet native speaker intuitions suggest a more nuanced pattern. In (3a), the procedural adjunct *tarife göre* ‘according to the recipe’ supports an inclusive interpretation, meaning that Ayşe may have baked the cake, but not according to the recipe. In (3b), with the non-creation verb *yıka-* ‘wash,’ the inclusive reading (‘she washed it, but not quickly’) is far less available. In (3c), with the creation verb *yaz-* ‘write,’ the inclusive option (‘she wrote the article, but not quickly’) resurfaces. These contrasts suggest that properties of the event predicate interact with adjunct recoverability, contradicting the AE-only prediction and motivating a closer experimental investigation.

This chapter builds on a line of work that began with a research proposal in Fincan (2023) and was first empirically implemented later that year, with results reported in Fincan (2024). Here I refine and expand those findings in light of recent debates, situating Turkish within the broader typology of ellipsis. I argue that Turkish object-gap constructions permit adjunct inclusion even in baseline antecedent + negated ellipsis contexts, a pattern that contradicts AE-only and polarity-only accounts and supports a restricted VSVPE conditioned by verb class.

The discussion proceeds as follows. Section 2 reviews the crosslinguistic literature and Turkish morphosyntax. Section 3 introduces the experiment. Section 4 reports the results, Section 5 discusses them, and Section 6 concludes.

Background

Literature on VSVPE and AE

VSVPE was first proposed in the late 1980s and early 1990s as a subtype of VP ellipsis. Classic cases were identified in languages such as Chinese (Huang, 1991), Japanese and Korean (Otani & Whitman, 1991), Hebrew (Doron, 1990), and Irish (McCloskey, 1991). Since then, VSVPE has been claimed for a range of languages, though its crosslinguistic availability remains contested. English does not exhibit VSVPE but provides a useful point of comparison through pseudogapping, where an auxiliary survives while the lexical verb is unpronounced and a remnant object surfaces, as in (4).

(4) John ate beans, and Mary did rice.

In (4), the auxiliary *did* licenses ellipsis while the remnant *rice* surfaces overtly. The verb *eat* is not repeated but is recovered from the antecedent clause. Turkish object gaps resemble pseudogapping in that both leave VP-internal material unexpressed and rely on movement out of an ellipsis site. This makes pseudogapping an important structural parallel, as it shows that the idea of material escaping an ellipsis domain has long-standing precedent in the literature. At the same time, Turkish object gaps differ crucially in retaining the lexical verb, which means pseudogapping serves here as a heuristic comparison rather than a one-to-one analysis.

Much of the early support for VSVPE came from Hebrew, where Goldberg (2005) introduced the Verbal Identity Requirement (VIR). According to this constraint, the stranded verb must match the antecedent in both root and derivational morphology. Evidence for VIR across languages is inconsistent. In Turkish, Us (2024) tested predicate adjective complements under ellipsis and reported that declaratives show sensitivity to VIR while polar questions permit mismatches. However, I believe his declaratives were often infelicitous even without ellipsis (e.g., collocationally

or in terms of discourse coherence), which complicates interpretation. Controlled testing of VIR in Turkish remains an open issue, so this chapter instead turns to adjunct inclusion as the central diagnostic.

The recent debate has been shaped by Landau (2018, 2020, 2023), who reanalyze putative VSVPE cases as argument ellipsis or, in many contexts, as deletion of PolP. His arguments are empirical and rest on the observation that, once discourse conditions are controlled, many adjunct-including readings disappear, leading to the prediction that baseline adjunct inclusion is impossible. On this view, recoverability of adjuncts is tied to polarity rather than to VP-deletion.

Crosslinguistic evidence has been invoked both to support and to challenge this reduction. In Russian, Gribanova (2017) shows that verb movement into Pol interacts with ellipsis to produce MaxElide effects.¹ Such effects cannot be derived from AE alone. In Persian, Shafei (2015) observes that ellipsis of complex predicates is acceptable in some mismatch configurations but sharply degraded in others. A verb-stranding account can explain this asymmetry by allowing the light verb to raise out of the deleted site. AE, by contrast, struggles to capture the contrast.

Other languages deepen the debate. Manetta (2019) argues that Hindi-Urdu shows restricted VSVPE under contrastive coordination and negation, whereas Landau (2023) contends that these are reducible to polarity ellipsis. Japanese has been analyzed similarly: Funakoshi (2016) takes adjunct recoverability under coordination and negation as evidence for VSVPE, while Landau again treats the facts as polarity ellipsis. Simpson (2023) expands the typology by comparing Hindi, Bangla, and Japanese, emphasizing that AE can explain argument drop but not adjunct recoverability, while VSVPE derives both. He concludes that the asymmetry between arguments and adjuncts cannot be captured without positing a verb-stranding mechanism.

Defenders of VSVPE therefore point to contexts where AE undergenerates, while Landau highlights cases where VSVPE overgenerates. This theoretical persistent clash makes Turkish especially important. If adjuncts are recoverable in baseline Turkish object gaps, this directly contra-

¹The MaxElide principle, introduced by Merchant (2008), states that if movement leaves a trace inside an ellipsis site, the largest possible constituent in the parallelism domain must be deleted. In Russian polarity-focus contexts, this requires deletion of TP rather than VP.

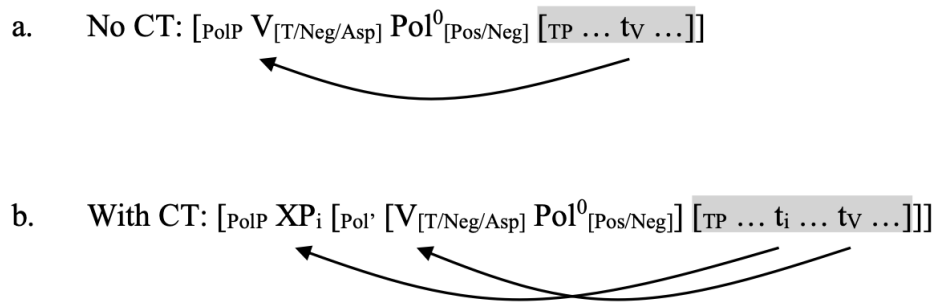
dicts Landau's prediction that such readings are impossible. It would mean that polarity ellipsis cannot account for the full typology and that Turkish requires a mechanism capable of deleting VP-internal material while stranding the verb.

Polarity Ellipsis and Alternatives to VSVPE

Because polarity ellipsis plays such a central role in Landau's program, it merits careful its own discussion. Landau (2020, 2023)'s proposal is that apparent verb-stranding is not VP-deletion but deletion of a higher clausal projection, PolP or TP, licensed by polarity features. In this analysis, the verb raises out of the ellipsis site, polarity determines the licensing domain, and adjunct recoverability is tied to polarity configurations rather than to VP structure. The surface outcome resembles VSVPE but the derivation is different.

As Landau (2023) unfolds, above TP sits a functional head Pol, which encodes clausal polarity. Inside TP, a head Σ realizes negation under agreement with Pol. The verb raises through T and can surface in Pol. An E-feature on Pol or T licenses deletion of their complement once agreement is established, which means the size of the ellipsis site is determined by polarity rather than by VP boundaries.

(5) Polarity ellipsis schemata



(Landau, 2023, p. 14)

In (5a), Pol licenses ellipsis of its TP complement, leaving only the verb. In (5b), a contrastive XP moves to [Spec,PolP] before deletion, so it remains overt together with the verb. Tense, agree-

ment, aspect, and even affixal negation can surface on the verb. These schemata explain why polarity ellipsis can mimic the output of AE or VSVPE.

The proposal's force comes from how polarity features regulate ellipsis. Ellipsis requires feature parallelism between antecedent and target. If the antecedent clause is positive but the ellipsis clause is negative, the Σ features do not match and deletion fails. This predicts that a baseline structure with an affirmative antecedent followed by a negated ellipsis clause cannot yield adjunct inclusion. Landau makes this explicit by stating that when Σ is valued [Neg], it mismatches the antecedent's [Pos]. When Σ is unvalued and must be valued by Pol, valuation occurs inside the deleted domain, bleeding agreement and producing a crash. Either way, baseline inclusion is blocked.

When polarity aligns across antecedent and ellipsis, adjunct inclusion becomes possible. If both are negative, speakers sometimes accept inclusive readings. Here, identity is established at the ΣP level rather than through VP-copying. Strongest support comes from contrastive contexts. When a contrastive XP moves into [Spec,PolP], Pol can license deletion of TP. Because agreement between Pol and Σ takes place before deletion, polarity reversal relative to the antecedent is also tolerated. The string that surfaces resembles VSVPE, but the underlying derivation is PolP-deletion licensed by discourse contrast. Landau therefore reinterprets cases of adjunct recoverability in languages like Japanese and Hindi as polarity ellipsis rather than verb-stranding, removing the need for VSVPE as an independent mechanism.

Verb-Final Languages and the Typology of Ellipsis

As previous sections illustrate, verb-final languages provide a key testing ground because their surface order makes the size of the ellipsis site transparent. Japanese has long been treated as an AE language. Sloppy readings and scope evidence support null arguments, and adjunct recoverability is absent in baseline structures (Oku, 1998; Saito, 2007; Şener & Takahashi, 2010). Later work shows that adjunct-inclusive readings do surface, but only under coordination or negation (Funakoshi, 2016). This restricted availability fits Landau's polarity ellipsis, as baseline inclusion is blocked, while polarity match or discourse contrast licenses deletion.

Hindi-Urdu has generated sharper controversy. Manetta (2019, 2021) argues that it allows restricted VSVPE, with verb evacuation from AspP licensed in contexts of contrastive coordination and negation. On this account, adjunct inclusion is genuine but context-sensitive. Landau (2020, 2023) disputes this, contending that once polarity ellipsis is factored in, no VP-deletion is needed. The disagreement centers on whether these cases truly diagnose verb-stranding or whether they can be captured by PolP-deletion.

Persian adds further complexity. Complex predicates built from a light verb and a lexical element show asymmetries in ellipsis acceptability. Shafiei (2015) takes these asymmetries to indicate verb-stranding, since the light verb can survive deletion. Landau (2020) instead treats them as AE constrained by discourse. Bangla introduces similar disagreements: argument drop is well accepted, but adjunct recoverability remains unsettled (Simpson, 2023). The broader typology is therefore not uniform.

Turkish enters as a particularly revealing case. Like Japanese and Hindi-Urdu, it is an SOV language with flexible adjunct placement. If Turkish were to permit adjunct inclusion in baseline configurations (an affirmative antecedent followed by a negated ellipsis clause, which is the sentence structure labeled as "impossible" by Landau, 2023) this would align it more closely with Hindi-Urdu and suggest that verb-stranding is not entirely absent from verb-final systems. It would also challenge the polarity-based account as it shows adjunct inclusion in the "impossible" structure. The typological significance of Turkish would then lie in showing that verb-final order does not by itself eliminate verb-stranding, and that any surviving VSVPE must be constrained not only by discourse polarity but also by properties of the verb and adjunct.

Turkish Linguistic Landscape

The broader typological debates set the stage for a closer look at Turkish itself. Turkish is an agglutinative, head-final language with rich verbal morphology and subject-predicate agreement. Agreement is realized by person-number suffixes on the verb, while objects do not participate in agreement (Kornfilt, 2003). The language is underlyingly SOV, but word order is highly flexible due to scrambling. Arguments and adjuncts may appear leftward, rightward, or in long-

distance positions, leading Turkish to be described as a “free word order” language (Aygen, 2001; Erguvanlı-Taylan, 1984; İnce, 2009; Kural, 1993, 1997; Özsoy, 2019; Şener & Takahashi, 2010). On one influential analysis, Turkish verbs raise overtly only as far as T° and not into C° (see Ulutaş, 2006, for discussion of NPI licensing, scope facts, and object shift). Under this view the verb surfaces in the functional domain where tense and agreement are realized, while the C-system is handled through feature percolation rather than verb movement. If this analysis is correct, then the combination of scrambling with V-to-T movement makes ellipsis sensitive to scope and argument structure, and the height of verb movement constrains whether adjuncts can be interpreted inside the ellipsis site.

Another central fact about Turkish is that it is analyzed as allowing null objects. Şener and Takahashi (2010) show that these cannot be analyzed as silent pronouns but must be elliptical. The argument comes from three diagnostics: sloppy readings, reflexive binding, and quantificational variability.

Consider sloppy readings in (6). If the null category were a pronoun, Condition B would block a sloppy interpretation, since a pronoun cannot be bound by the local subject. Yet Turkish allows both strict and sloppy readings. Ahmet can be understood as yelling at Ali’s friend (strict) or at his own friend (sloppy). The sloppy reading indicates that the null category copies structure from the antecedent, as expected under ellipsis.

- (6) a. Ali [pro arkadaş-ı]-na sarıl-dı.
 Ali pro friend-POSS-DAT hug-PST
 ‘Ali hugged his friend.’
- b. Ahmet-se *e* bağırdı.
 Ahmet-however *e* yell-PST
 ‘Ahmet, however, yelled at *e*.’
 = Ahmet yelled at Ali’s friend.
 = Ahmet yelled at Ahmet’s friend.

Reflexive binding supports the same conclusion. In (7), the reflexive *kendi kendine* must be locally bound. When the null object appears in the continuation, Ali can be interpreted as talking to himself. A pronominal analysis would not permit this binding relation, but ellipsis does, because the elided structure contains the reflexive.

- (7) a. Kim kendi kendi-ne konuş-uyor?
 who self self-DAT talk-PROG
 ‘Who is talking to themself?’
 b. Ali *e* konuş-uyor.
 Ali *e* talk-PROG
 ‘Ali is talking to *e*.’
 = Ali is talking to himself.

Quantificational variability provides a third diagnostic. In (8), Ali draws two pictures, and in the continuation Ayşe paints *e*. The sentence is ambiguous: she may paint the same two pictures or two different ones. This ambiguity arises because the elided object is interpreted via copying from the antecedent, not because of pronominal reference, which would force a single reading.

- (8) a. Ali iki resim çiz-di.
 Ali two picture draw-PST
 ‘Ali drew two pictures.’
 b. Ayşe-se *e* boya-dı.
 Ayşe-however paint-PST
 ‘Ayşe, however, painted *e*.’
 = the same two pictures, or two different ones.

From these diagnostics, Şener and Takahashi (2010) conclude that Turkish null objects are elliptical rather than pronominal. The next step is to determine which ellipsis mechanism is responsible. To this end, they turn to the adjunct inclusion test, which diagnoses whether ellipsis targets only the argument or a larger VP domain. If the ellipsis site includes the VP, an adjunct contained

inside it should be able to survive; if the site is limited to the object, adjuncts will not be recoverable.

Their evidence comes from sentences like (9). In the antecedent clause, the problem is solved quickly. If the object gap in the second clause involved VP-ellipsis with verb-stranding, the adverb *hızla* ‘quickly’ would fall inside the deleted constituent and should be interpreted in the ellipsis clause as well. In that case, the sentence ought to allow the reading “Phylis did not solve the problem quickly”.

- (9) a. Can sorun-u hızla çöz-dü.
 Can problem-ACC quickly solve-PST
 ‘John solved the problem quickly.’
- b. Filiz-se *e* çöz-me-di.
 Phylis-however *e* solve-NEG-PST
 ‘*Lit.* Phylis, however, did not solve *e*.’
 = Phylis did not solve the problem.
 ≠ Phylis did not solve the problem quickly.

(Şener & Takahashi, 2010, p. 90)

Yet Şener and Takahashi (2010) report that this reading is unavailable, and the only possible interpretation is the adjunct-excluding one. On this basis, they conclude that the ellipsis site cannot be VP; it must be the argument position alone, consistent with AE.

At first sight, (9) appears to decisively favor AE. However, the choice of adjunct is problematic. As shown by Erguvanlı-Taylan (1984), Temürcü (2001), and Üntak Tarhan (2006), Turkish manner adverbs fall into distinct morphosyntactic classes, and *hızla* does not belong cleanly to either. This raises the possibility that the unavailability of the adjunct-including reading is tied to the lexical properties of *hızla* rather than to a structural ban on adjunct recoverability.

In addition, Şener and Takahashi (2010) did not consider other types of adjuncts, such as procedural ones (*-e göre* ‘according to X’), which play a central role crosslinguistically in licensing

inclusion readings. Native speaker intuitions suggest that such adjuncts can sometimes be interpreted under ellipsis, particularly with creation verbs, which complicates the AE-only conclusion. For instance, in (10), the procedural adjunct supports an inclusive interpretation: Eda baked, but not according to the recipe.

- (10) Burak kek-i tarif-e göre yap-mış. Eda yap-ma-mış. O
 Burak cake-ACC recipe-according to make-PST.EVID Eda make-NEG-PST.EVID
 yüzden Eda'nın keki kabar-ma-mış.

‘Burak made the cake according to the recipe. Eda didn’t. That’s why Eda’s cake did not rise.’

As reviewed in Section 2 (§2.1–§2.2), AE limits deletion to arguments and Polarity Ellipsis requires feature parallelism (affirmative–negative baselines block deletion and adjunct identity).

(10) instantiates that “impossible” baseline (Landau, 2023) yet permits adjunct inclusion, directly contradicting AE+Polarity. We therefore test this configuration systematically in Section 3.

Finally, Turkish has productive light-verb constructions (e.g., *yardım et-* ‘help,’ *dans et-* ‘dance’), similar to Persian, where the VSVPE vs. AE debate is also unsettled (Shafiei, 2015). Whether Turkish light verbs behave like simplex verbs under ellipsis or introduce new structural layers remains open, and future research will need to address this domain.

Experiment

This study directly tests one of the core predictions of Şener and Takahashi (2010), namely that adjunct inclusion is categorically excluded in Turkish object gaps and that such a result would support an AE-only analysis. My hypothesis, by contrast, is that adjunct inclusion is possible. The experimental materials in fact covered a broader set of conditions than I report here. In addition to according to ‘x’ adjuncts and simple manner adverbs, I also tested complex manner adverbs, as well as structures with focused and non-focused adverbs. The design further included crosslinguistic probes, drawing on Japanese and Hindi examples involving rich contrastive con-

texts, coordination with *but*, and negated antecedents. For reasons of clarity and scope, the present report concentrates only on *-e göre* ‘according to X’ adjuncts and simple manner adverbs such as *hızlı* ‘quickly.’ This choice is motivated by two considerations. First, the syntactic position of simple manner adverbs is well established as VP-internal (Üntak Tarhan, 2006), which makes them a stable diagnostic. Second, ‘according to X’ adjuncts have been widely used in crosslinguistic debates about ellipsis, making them a natural point of comparison. The other conditions, while informative, are reserved for later work.

Design

The experiment manipulated two factors. The first factor (Adjunct Type) compared procedural adjuncts (*-e göre* ‘according to X’) with simple manner adverbs such as *hızlı* ‘quickly.’ The second factor (Continuation Type) compared two possible continuations, illustrated in (11).

- (11) a. Burak kek-i tarif-e göre yap-mış. Eda yap-ma-mış.
 Burak cake-ACC recipe-according to make-PST.EVID Eda make-NEG-PST.EVID
 O yüzden Eda’nın keki kabarmamış.

Burak made the cake according to the recipe. Eda didn’t (make). That’s why Eda’s cake did not rise.

- b. Burak kek-i tarif-e göre yap-mış. Eda yap-ma-mış.
 Burak cake-ACC recipe-according to make-PST.EVID Eda make-NEG-PST.EVID
 Bu hafta kek yapmadığı için haftaya kek yapma sırası Eda’daymış.

Burak made the cake according to the recipe. Eda didn’t (make). Since Eda didn’t make a cake this week, it was Eda’s turn next week.

In the inclusion condition (2A), the third sentence forced an interpretation where the adjunct had to be recovered in the ellipsis, as in (11a). In the exclusion condition (2B), the third sentence forced an interpretation where the adjunct could not be recovered, as in (11b). In both conditions,

the antecedent clause and the ellipsis clause were identical; only the continuation sentence varied. Participants rated the naturalness of each three-sentence sequence on a 7-point Likert scale (1 = least natural, 7 = most natural).

A native-speaker consultant reviewed the stimuli for naturalness and ambiguity; subsequent revisions incorporated supervisor feedback. Stimuli were constructed to use the “baseline” configuration (affirmative antecedent + negated ellipsis) so that any adjunct-inclusion effect would *not* rely on coordination, rich contrast, or a negated antecedent.

Design confound and corrective Phase 2

During analysis of Phase 1, we discovered that procedural adjuncts were paired with creation verbs, while SMAs were paired with non-creation verbs. To address this potential confound, Phase 2 re-tested SMAs with creation verbs only.

Table 1: Verb–adjunct pairings by phase (schematic).

Phase	Adjunct type	Verb class (examples)
Phase 1	Procedural (<i>-e göre</i>)	Creation (e.g., <i>yap-</i> ‘make’)
Phase 1	Simple manner (SMA)	Non-creation (e.g., <i>yıka-</i> ‘wash’)
Phase 2	Simple manner (SMA)	Creation (e.g., <i>yaz-</i> ‘write’)

Procedure

The task was implemented in PsychoPy and deployed via Pavlovia/Google Forms. Four lists (Phase 1) counterbalanced lexicalizations across conditions (32 items per list: 16 test, 16 fillers). Items were randomized per participant.

Sixty native speakers of Turkish (born/raised in Turkey; L1 confirmed in screening) completed Phase 1; two additional recruits did not complete the task. Phase 2 used the same methodology with a smaller set (8 items: 4 test, 4 fillers); 60 participants took part across two lists (29/31). Participants rated each narrative on a 7-point Likert scale. Following common practice, we analyze these ratings as approximately continuous (e.g., Norman, 2010). To check robustness, we also fit cumulative-link (ordinal) models and inspected by-item means; the pattern of results was

unchanged.

Exclusion criteria were preregistered: (i) failure on an attention check, or (ii) response-time outliers with median absolute deviation (MAD) > 2.5 relative to a participant's own distribution.

This led to 3 exclusions in Phase 1. No participants were excluded in Phase 2. Analyses with and without exclusions yielded the same qualitative pattern.

Results

Phase 1

Ratings were analyzed with a linear mixed-effects model predicting naturalness from ADJUNCT-TYPE (1A vs. 1B), CONTINUATION (2B vs. 2A), and their interaction, with random intercepts for participants. Three participants met the exclusion criteria (above), leaving $n = 57$.

Table 2: Phase 1: Cell means (AdjunctType $\times$ Continuation). 7-point Likert.

AdjunctType	Continuation	Mean	SD
1A	Exclusion (2B)	4.93	1.99
1A	Inclusion (2A)	4.98	2.36
1B	Exclusion (2B)	4.53	2.18
1B	Inclusion (2A)	3.44	2.38

Descriptively, 1A shows parity between Inclusion and Exclusion ($\Delta = +0.05$), while 1B shows an exclusion advantage ($\Delta = -1.09$). For interpretability, we also report mean differences (Inclusion – Exclusion) with 95% CIs: 1A: $\Delta = +0.05$, 95% CI $[-0.72, +0.79]$; 1B: $\Delta = -1.09$, 95% CI $[-1.94, -0.13]$.

Table 3: Phase 1: Linear mixed-effects estimates (reference: 1A–2B).

Predictor	Estimate (β)	95% CI	<i>p</i>
Intercept (1A–2B)	4.93	[4.4, 5.5]	< .001
AdjunctType (1B vs 1A)	–0.40	[–1.0, 0.2]	.18
Continuation (2A vs 2B)	+0.05	[–0.6, 0.7]	.89
Interaction (1B×2A)	–1.10	[–1.8, –0.4]	.002

The interaction is reliably negative: the Inclusion–Exclusion difference is significantly larger (more negative) in 1B than 1A, indicating an Inclusion penalty only in the 1B configuration.

Phase 2

Phase 2 re-tested the 1B configuration using *creation* verbs. No participants were excluded. We fit the same participant-intercepts model. Robustness checks with ordinal regression and by-item means showed the same null effect.

Table 4: Phase 2 (1B with creation verbs): Cell means. 7-point Likert.

Continuation	Mean	SD	N
Exclusion (2B)	4.97	1.93	60
Inclusion (2A)	5.23	2.04	60

Here Inclusion slightly exceeds Exclusion ($\Delta = +0.26$, 95% CI [–0.37, +0.90])...

Table 5: Phase 2: Linear mixed-effects estimate for Inclusion (2A) vs. Exclusion (2B).

Predictor	Estimate (β)	SE	95% CI	<i>p</i>
Intercept (2B)	4.97	0.26	[4.46, 5.47]	< .001
Continuation (2A vs 2B)	+0.27	0.32	[–0.36, 0.89]	.40

In sum, Phase 1 shows an Inclusion penalty for SMA with non-creation verbs (1B), but Phase 2 removes that penalty when SMAs are paired with creation verbs.

Discussion

The overall pattern is incompatible with an AE-only analysis. As argued in Section 2.1, AE predicts categorical adjunct exclusion, yet both procedural adjuncts and SMAs (when paired with creation verbs) allow inclusion. Given independent evidence that SMAs are VP-internal (Üntak Tarhan, 2006), their recoverability suggests that the VP itself is targeted by ellipsis, leaving the verb stranded.

Phase 1 showed a sharp asymmetry: inclusion was penalized in the 1B condition but not in 1A. Phase 2 re-tested 1B with creation verbs to address the confound that Phase 1 always paired 1B with non-creation verbs. In this follow-up, the exclusion penalty did not persist: ratings for Inclusion and Exclusion were roughly comparable. This suggests that the earlier bias may have been tied to verb choice rather than adjunct type per se. In other words, verb class may modulate adjunct recoverability, with creation predicates tending to permit inclusion and result-state predicates constraining it. A more systematic test of verb type is needed to confirm this possibility. Polarity ellipsis cannot capture this profile. The stimuli avoided contexts that typically facilitate PolP-deletion (coordination, negated antecedents, rich contrast), yet adjunct inclusion still emerged in the baseline antecedent + negated-ellipsis configuration that Landau (2023) classifies as “impossible.” Moreover, the effect is not discourse-general but appears to be sensitive to event structure. This kind of granularity is unexpected under polarity ellipsis but is consistent with accounts where event structure conditions VP-deletion.

Typologically, Turkish therefore enlarges the space of verb-final languages where VSVPE must be recognized. Unlike Japanese, where adjunct inclusion surfaces mainly in contrastive or negative contexts, Turkish permits it in baseline environments. In this respect it patterns closer to Hindi’s restricted VSVPE (Manetta, 2019), showing that verb-final order does not block verb-stranding. In short, adjunct inclusion in Turkish object gaps is possible without discourse scaffolding, but its availability seems to be modulated by verb type. AE-only and polarity-only accounts cannot capture this distribution. The evidence supports retaining VSVPE in the grammar of Turkish, while future work should test more systematically how event semantics constrains adjunct recoverabil-

ity.²

Conclusion

This chapter isolates a core empirical fact about Turkish object gaps, namely that adjunct interpretation can be recovered under ellipsis even in baseline, non-contrastive environments. By designing around discourse configurations that typically aid recoverability, the study shows that the effect does not depend on discourse scaffolding. The resulting profile forces a grammatical account in which predicate-internal material can be silenced while the verb survives. In other words, a verb-stranding option must be available in Turkish. In doing so, the findings directly contradict the AE-only position of Şener and Takahashi (2010), which predicted categorical exclusion of adjuncts.

The contribution is twofold. Empirically, the data identify a dependency between adjunct recoverability and event structure, with creation and scalar-change predicates patterning differently. Theoretical accounts that derive silent-adjunct readings solely from polarity licensing struggle to predict this fine-grained sensitivity to verb class and adjunct type. Conceptually, the findings narrow the explanatory space: an AE-only view is insufficient, while a polarity-only view under-generates; a restricted VSVPE that is conditioned by event semantics emerges as the most plausible architecture.

Typologically, Turkish expands the landscape for verb-final languages. It aligns with systems that require a verb-stranding mechanism in at least limited domains, showing that head-final order does not preclude VP-targeted deletion. In doing so, Turkish helps refine crosslinguistic diagnostics: what matters is not word order per se, but how verb class, adjunct type, and clause structure conspire to license ellipsis.

Several avenues follow directly. First, a compositional account of creation vs. non-creation predicates should be integrated with the syntax of ellipsis, making explicit which layers of event struc-

²One might propose that certain adjuncts are in fact selected arguments of particular verbs. While this is in principle a possible line of analysis, the burden of proof lies with those who advance it. Such a claim would require independent evidence that these putative adjuncts behave like arguments across contexts (e.g., in selectional restrictions, argument structure diagnostics, or binding). Without such evidence, the adjunct-as-argument view remains speculative and does not undercut the present argument.

ture are targeted by deletion. Second, controlled tests in polarity-focus configurations can probe whether Turkish patterns like other languages where discourse and polarity independently license ellipsis. Third, extending the paradigm to complex manner adverbs and light-verb constructions will reveal whether the same constraints persist across more articulated predicate structures. Finally, individual-difference modeling and prosodic manipulation tasks could determine whether small between-speaker shifts reflect grammatical optionality or processing preferences.

In sum, Turkish object gaps require a grammar that admits verb-stranding ellipsis moderated by event semantics. Recognizing this restricted VSVPE option not only captures the Turkish pattern but also sharpens the crosslinguistic typology of ellipsis, clarifying where discourse, argument deletion, and predicate-internal structure each do explanatory work.

References

- Aygen, G. (2001). *Finiteness, clause structure, and case* [Doctoral dissertation]. Harvard University.
- Doron, E. (1990). *V-movement and VP-ellipsis* [Manuscript, Hebrew University of Jerusalem].
- Erguvanlı-Taylan, E. (1984). *The function of word order in Turkish grammar*. University of California Press.
- Fincan, Y. (2023). *Object gaps in Turkish: Argument ellipsis or VSVPE?* [Unpublished Manuscript, Eberhard Karls Universität Tübingen].
- Fincan, Y. (2024). *Verb phrase ellipsis beyond English: A brief study of object gaps in Turkish* [Master's thesis]. University of Tübingen, Department of English.
- Funakoshi, K. (2016). Verb-stranding verb phrase ellipsis in Japanese. *Journal of East Asian Linguistics*, 25(2), 113–142.
- Goldberg, L. (2005). *Verb-stranding VP ellipsis: A cross-linguistic study* [Doctoral dissertation]. McGill University.
- Gribanova, V. (2013). Verb-stranding verb phrase ellipsis and the structure of the Russian verbal complex. *Natural Language and Linguistic Theory*, 31(1), 91–136.

- Gribanova, V. (2017). Head movement and ellipsis in the expression of Russian polarity focus. *Natural Language & Linguistic Theory*, 35(4), 1079–1121.
- Huang, C.-T. J. (1991). Remarks on the status of the null object. In R. Freidin (Ed.), *Principles and parameters in comparative grammar* (pp. 56–76). MIT Press.
- İnce, A. (2009). *Dimensions of ellipsis: Investigations in Turkish* [Doctoral dissertation]. University of Maryland, College Park.
- Kornfilt, J. (2003). Scrambling, subscrambling, and case in Turkish. In S. Karimi (Ed.), *Word order and scrambling* (pp. 125–155). Blackwell.
- Kural, M. (1993). Scrambling and mixed positions in Turkish. *Proceedings of the North East Linguistic Society*, 22, 259–271.
- Kural, M. (1997). Postverbal constituents in Turkish and the linear correspondence axiom. *Linguistic Inquiry*, 28(3), 498–519.
- Landau, I. (2018). Missing objects in Hebrew: Argument ellipsis, not VP ellipsis. *Glossa: A Journal of General Linguistics*, 3(1), 1–33.
- Landau, I. (2020). On the nonexistence of verb-stranding VP-ellipsis. *Linguistic Inquiry*, 51(2), 341–365.
- Landau, I. (2023). More doubts on verb-stranding VP ellipsis: Reply to simpson 2023. *Syntax*, 26(4), 449–470.
- Manetta, E. (2019). Verb-phrase ellipsis and complex predicates in Hindi-Urdu. *Natural Language and Linguistic Theory*, 37(3), 915–953.
- Manetta, E. (2021). Verb-second and the verb-stranding verb phrase ellipsis debate. *Glossa: A Journal of General Linguistics*, 6(1), 77.
- McCloskey, J. (1991). Clause structure, ellipsis, and proper government in Irish. *Lingua*, 85, 259–302.
- Merchant, J. (2008). Variable island repair under ellipsis. In K. Johnson (Ed.), *Topics in ellipsis* (pp. 132–153). Cambridge University Press.

- Norman, G. (2010). Likert scales, levels of measurement and the “laws” of statistics. *Advances in Health Sciences Education*, 15(5), 625–632.
- Oku, S. (1998). *A theory of selection and reconstruction in the minimalist perspective* [Doctoral dissertation]. University of Connecticut.
- Otani, K., & Whitman, J. (1991). V-raising and VP-ellipsis. *Linguistic Inquiry*, 22, 345–358.
- Özsoy, A. S. (2019). Derivation of fronting in Turkish. In A. S. Özsoy (Ed.), *Word order in Turkish* (pp. 139–177, Vol. 97). Springer.
- Park, M.-K. (1997). The syntax of VP ellipsis in Korean. *Language Research*, 33, 629–648.
- Saito, M. (2007). Notes on East Asian argument ellipsis. *Language Research*, 43(2), 203–227.
- Şener, S., & Takahashi, D. (2010). Ellipsis of arguments in Japanese and Turkish. *Nanzan Linguistics*, 6, 79–99.
- Shafiei, N. (2015). Ellipsis in Persian complex predicates: VVPE or something else? *Proceedings of the Annual Conference of the Canadian Linguistic Association*.
- Simpson, A. (2023). In defense of verb-stranding VP ellipsis. *Syntax*, 26, 431–448.
- Temürcü, C. (2001). *Word order variations in Turkish: Evidence from binding and scope* [Master’s thesis]. Middle East Technical University.
- Ulutaş, S. (2006). *Verb movement and feature percolation: Evidence from Turkish* [Master’s thesis]. Boğaziçi University.
- Üntak Tarhan, F. A. (2006). *Topics in the syntax–phonology interface in Turkish: Sentential stress and phases* [Master’s thesis]. Boğaziçi University.
- Us, E. (2024). Verb-stranding VP ellipsis in Turkish. *Proceedings of the 18th Workshop on Altaic Formal Linguistics (WAFL 18)*.